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Intent-Aware Deception System for Real-Time Attacker Profiling

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> **Purpose of this document.** This is a complete, section-by-section source manuscript for an IEEE-format conference paper. Everything you need is here -- abstract, related work, architecture, methodology, real experimental results, discussion, limitations, and references. **Every quantitative claim is real** and traceable to `backend/app/ml/artifacts/metrics.json` or to the cited source file. Copy sections directly into the IEEE two-column template; the tables are already formatted.

> Suggested venue framing: a systems + applied-ML security paper (honeypots / intrusion analysis / applied machine learning).

Title

Intent-Aware Deception: Real-Time Attacker Profiling via Hybrid Rule-ML Classification of Honeypot Command Streams

Alternative shorter title: **Intent-Aware Deception System for Real-Time Attacker Profiling**

Abstract

Traditional honeypots excel at *capturing* attacker activity but stop short of *interpreting* it: they log commands without inferring the adversary's underlying goal. We present an **intent-aware deception system** that transforms raw honeypot telemetry into real-time, explainable attacker profiles. Captured shell commands from an isolated medium-interaction honeypot are streamed through an asynchronous pipeline that normalises each command, extracts lexical features, and classifies it into one of **ten MITRE ATT&CK-aligned intent classes** using a **hybrid classifier**: a deterministic rule engine, a calibrated machine-learning core, and an optional large-language-model semantic layer, arbitrated by explicit confidence precedence. Per-session and per-attacker profiles are assembled into multi-stage attack *trajectories*, and each session receives an **explainable risk score (0-100)** from an additive engine combining intent weights and behavioural boosters. The ML core -- a calibrated linear support-vector machine over a combined word- and character-n-gram TF-IDF feature space -- achieves **94.49 % accuracy** and **0.9229 macro-F1** on a leakage-controlled held-out set of 272 commands; a deliberately stricter *leave-families-out* protocol yields **67.87 % accuracy**, reported explicitly to characterise generalisation honestly. The full system -- capture, classification, scoring, persistence, REST/WebSocket API, and a live analyst dashboard -- runs end-to-end in real time and is reproducible from a fixed seed. We argue that fusing deterministic rules, calibrated ML, and an optional LLM yields profiling that is simultaneously accurate, real-time, and auditable -- a combination none of the three achieves alone.

Index Terms -- honeypot, cyber deception, attacker profiling, intent classification, MITRE ATT&CK, machine learning, text classification, threat intelligence, explainable security, support vector machine.

I. Introduction

Honeypots -- decoy systems with no legitimate purpose -- are a cornerstone of cyber deception: because no benign user should ever touch them, *every* interaction is, by definition, suspicious [1], [2]. Medium-interaction honeypots such as Cowrie [3] emulate a Unix shell, enticing attackers to reveal their tooling and tradecraft while safely containing them. However, the dominant output of such systems is a *log*: a chronological record of commands that a human analyst must still read and interpret. At scale, this interpretation burden is the bottleneck.

The gap we address is **semantic**: existing honeypots tell you *what* an attacker typed, but not *why*. Knowing that an attacker ran `cat /etc/shadow` followed by `wget http://.../x.sh` and then `rm -rf /var/log/auth.log` is far more actionable when the system recognises this as **credential access -> execution -> defensive cleanup** -- a coherent attack progression -- and raises a scored alert automatically.

This paper presents an **intent-aware deception system** that closes that gap in real time. Our contributions are:

1. **A 10-class, MITRE ATT&CK-aligned intent taxonomy** for shell commands, with a deterministic mapping from each intent to an ATT&CK tactic/technique and to a kill-chain *attack stage* (Section IV).

2. **A hybrid classifier** that arbitrates a deterministic rule engine, a calibrated ML core, and an optional LLM semantic layer by explicit confidence precedence, recording the deciding source for every event so that every classification is auditable (Section VI).

3. **An explainable additive risk-scoring engine** (0-100) that combines per-intent weights with behavioural boosters and maps to four severity tiers with automatic alerting (Section VII).

4. **A complete, real-time, reproducible implementation** -- asynchronous capture pipeline, MongoDB persistence, FastAPI REST/WebSocket API, and a live React analyst dashboard -- with the honeypot strictly isolated so **no captured command is ever executed** on the host or backend (Sections IX, and the safety discussion in XII).

5. **An honest experimental evaluation** with leakage control and a stricter leave-families-out protocol, reporting both the favourable and the conservative numbers (Sections X-XI).

II. Related Work

Honeypots and deception. Spitzner [1] and Provos and Holz [2] established honeypots as instruments for observing attacker behaviour; Cowrie [3] is the de-facto medium-interaction SSH/ Telnet honeypot, emulating a shell and logging structured JSON events. These systems are excellent *sensors* but are typically paired with manual or signature-only analysis.

Command and intrusion classification. Text-classification techniques over command lines and system logs are well established; TF-IDF vector-space representations [4] with linear classifiers such as support-vector machines [5] remain strong, interpretable baselines for short, high-dimensional sparse text. Character-n-gram features are known to be robust to the lexical variation and obfuscation typical of malicious inputs.

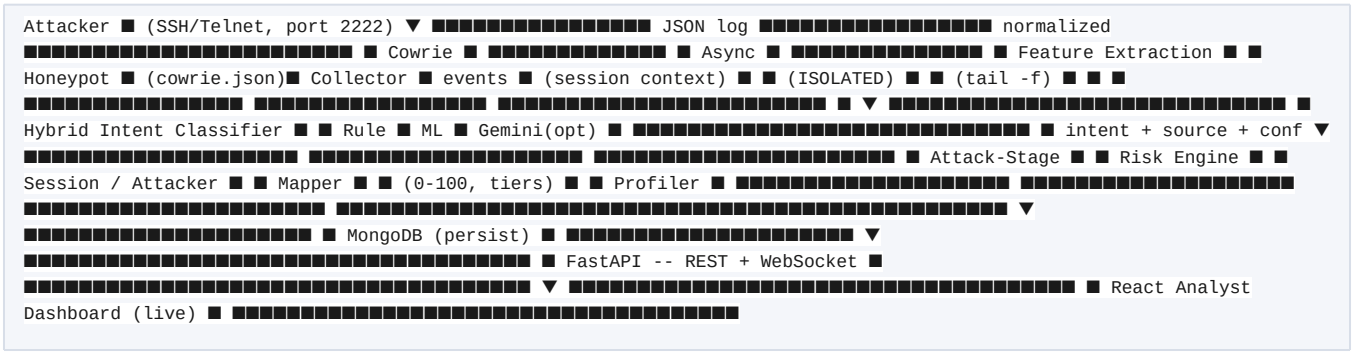
Adversary modelling. MITRE ATT&CK [6] provides a standard taxonomy of adversary tactics and techniques and has become the lingua franca for describing intrusions. Mapping observed behaviour onto ATT&CK tactics enables shared vocabulary and downstream automation.

LLMs for security triage. Recent work applies large language models to security-log summarisation and triage. LLMs offer strong semantic generalisation but introduce latency, cost, non-determinism, and availability dependencies -- properties ill-suited to a *mandatory* real-time path.

Our position. Rather than choosing one paradigm, we *fuse* them: deterministic rules give precision and auditability on known signatures; a calibrated ML model gives generalisation and probabilistic confidence; and an LLM is an *optional* fallback for genuinely novel or ambiguous commands. The novelty is not any single classifier but the **arbitration architecture** and its integration into a real-time, explainable, end-to-end profiling system.

III. System Architecture

The system is a streaming pipeline from honeypot capture to analyst dashboard:



Data flow. (1) The **honeypot** captures a login/command and appends a JSON event to its log. (2) An **asynchronous collector** tails the log and emits normalised events without blocking. (3) **Feature extraction** attaches session context (command counts, login failures, timing). (4) The **hybrid classifier** assigns an intent, a deciding source, and a confidence. (5) The **attack-stage mapper** places the intent on the kill chain; the **risk engine** scores the session; the **profiler** aggregates per-session and per-attacker state. (6) Everything is **persisted** to MongoDB and (7) exposed via a **FastAPI** REST + WebSocket API to (8) a **live React dashboard**.

Design principles. (i) *ML is mandatory, the LLM is optional* -- the system starts and operates fully without any API key. (ii) *Every decision is auditable* -- each event stores its deciding source and evidence. (iii) *Strict isolation* -- the honeypot is the only component that touches attacker input, and it only emulates a shell.

IV. Intent Taxonomy and MITRE Mapping

We define **ten intent classes**, each mapped deterministically to a MITRE ATT&CK tactic/ technique and to a kill-chain **attack stage**. The mapping is a static table in code (`taxonomy.py`), so intent -> tactic -> stage is reproducible, not inferred.

Intent class	MITRE tactic	Representative technique(s)	Attack stage
RECONNAISSANCE	Reconnaissance (TA0043)	Active Scanning (T1595); Network Service Discovery (T1046)	RECONNAISSANCE
DISCOVERY	Discovery (TA0007)	System Information (T1082); File/Dir Discovery (T1083); Process Discovery (T1057)	DISCOVERY
CREDENTIAL_ACCESS	Credential Access (TA0006)	OS Credential Dumping (T1003); Unsecured Credentials (T1552)	CREDENTIAL_ACCESS
EXECUTION	Execution (TA0002)	Command & Scripting Interpreter (T1059)	EXECUTION
PRIVILEGE_ESCALATION	Privilege Escalation (TA0004)	Abuse Elevation Control (T1548); Exploitation for Priv-Esc (T1068)	PRIVILEGE_ESCALATION
PERSISTENCE	Persistence (TA0003)	Account Manipulation (T1098); Scheduled Task (T1053); Create Account (T1136)	PERSISTENCE
EXPLOITATION	Execution / Initial Access	Exploit Public-Facing Application (T1190)	EXECUTION
EXFILTRATION	Exfiltration (TA0010)	Exfil over C2 (T1041); Exfil over Alt. Protocol (T1048)	EXFILTRATION
CLEANUP	Defense Evasion (TA0005)	Indicator Removal -- clear logs/history (T1070)	CLEANUP
OTHER	--	(no confident tactic)	UNKNOWN

Note two deliberate mapping choices: **EXPLOITATION** maps to the **EXECUTION** stage (an exploit is a means of code execution), and **OTHER** maps to **UNKNOWN** (we do not force a kill-chain position onto unclassified input). The full ordered attack-stage enumeration is: `INITIAL_ACCESS, RECONNAISSANCE, DISCOVERY, CREDENTIAL_ACCESS, EXECUTION, PRIVILEGE_ESCALATION, PERSISTENCE, EXPLOITATION, EXFILTRATION, CLEANUP` (with `UNKNOWN` as a fallback).

V. Machine-Learning Methodology

(This section is condensed; the companion document `ML_MODEL_ANALYSIS.md` carries every hyperparameter, the full confusion matrix, and the reproducibility recipe.)

A. Task

Supervised multi-class classification of a single shell-command string into one of the ten intent classes. The classifier consumes **only the raw command text**; behavioural session features are reserved for the risk engine (Section VII).

B. Dataset

The corpus is generated programmatically from **242 curated command "template families"**. Each family declares an intent and a parameterised template (or explicit variants); templates are expanded by substituting from value pools (IPs, ports, paths, payload names, CVE IDs, ...), capped at 14 variants per family, then de-duplicated. After removing 5 exact duplicates the final corpus is **1360 unique commands** across 10 classes (Table I). Labels are assigned *by construction* (each command inherits its family's intent), and the `family_id` is retained as a group key for the robustness split.

TABLE I -- Class distribution (samples / families).

Intent	Samples	Families
RECONNAISSANCE	80	28
DISCOVERY	134	35
CREDENTIAL_ACCESS	121	27

EXECUTION	318	24
PRIVILEGE_ESCALATION	57	30
PERSISTENCE	162	20
EXPLOITATION	100	18
EXFILTRATION	184	18
CLEANUP	87	18
OTHER	117	24
Total	**1360**	**242**

C. Features

A scikit-learn `FeatureUnion` of two TF-IDF representations: a **word** vectoriser (`analyzer=word`, whitespace tokenisation `(?u)\S+`, n-grams (1,2), `sublinear_tf`) and a **character** vectoriser (`analyzer=char_wb`, n-grams (3,6), `sublinear_tf`). The whitespace tokeniser deliberately keeps flags, paths, and URLs intact (`-O`, `/etc/shadow`), and the character n-grams capture sub-token cues (flag fragments, path fragments, obfuscation) that a word-only model misses -- the single most important design choice for the shell-command domain.

D. Model selection

Three candidates over the identical feature space were compared by **4-fold stratified cross-validation on the training split only**, scoring macro-F1 (Table II). The winner is a **calibrated linear SVM**: `LinearSVC(C=1.0, class_weight="balanced")` wrapped in `CalibratedClassifierCV(method="sigmoid", cv=3)`. Calibration (Platt scaling [7]) converts SVM margins into probabilities that the runtime hybrid layer thresholds on. (A RandomForest was evaluated during development and dropped for poor generalisation and calibration.)

TABLE II -- Model selection (4-fold CV macro-F1 on the training split).

Candidate	Estimator	CV macro-F1
linsvc_calibrated (selected)	Calibrated LinearSVC	**0.8539 +/- 0.0181**
logreg	LogisticRegression (C=3.0)	0.8482 +/- 0.0172
complement_nb	ComplementNB (α=0.3)	0.8146 +/- 0.0282

E. Protocol and leakage control

A stratified **80/20** split (1088 train / 272 test, `random_state=1337`). The code **asserts that no identical command string appears in both train and test** -- this leakage control is why the headline accuracy is a realistic 94 %, not an inflated 100 %. The shipped artefact is then refit on all 1360 samples and serialised with metadata (classes, intent->stage map, model name, sklearn version, training timestamp); loading is lazy and degrades gracefully to rules+LLM if the artefact is absent.

VI. Hybrid Classification

At inference, three sources are arbitrated by explicit **confidence precedence** (`hybrid_classifier.py`):

```
1. Strong RULE : rule confidence >= 0.90 and label ≠ OTHER -> source RULE 2. Confident ML : calibrated ML probability >= 0.40 -> source ML (HYBRID if rule agrees) 3. Gemini (opt.) : consulted only when 1 and 2 are inconclusive -> source GEMINI 4. Best-available: highest-confidence non-OTHER among rule/ML -> source RULE/ML 5. Fallback : nothing confident (>= 0.10) -> OTHER, source FALLBACK
```

- The **rule engine** comprises 13 rules over 43 curated regular-expression patterns, each with an attached confidence; a strong (>=0.90) non-OTHER match is authoritative -- deterministic, instant, and auditable.
- The **ML core** decides when no strong rule fires but its calibrated probability is >= 0.40; when the rule *agrees* with the ML label the event is recorded as **HYBRID** (both concur), otherwise **ML**.
- **Gemini** (`gemini-2.5-flash`, via the optional `google-genai` SDK) is consulted **only** when neither rule nor ML is decisive -- i.e. genuinely novel or ambiguous input. Its output is gated (accepted at >=0.85, considered at >=0.70) and its input is sanitised. It is never on the hot path for known attacks, and the system runs fully without it.

- Every event stores its **deciding source** (`RULE`/`ML`/`HYBRID`/`GEMINI`/`FALLBACK`) and, for ML/HYBRID, the calibrated confidence -- surfaced on the dashboard so no decision is a black box.

This arbitration is the core systems contribution: it combines the *precision and auditability* of rules, the *generalisation and calibrated confidence* of ML, and the *semantic reach* of an LLM, while keeping ML mandatory and the LLM optional.

VII. Explainable Risk Scoring

Each session receives an integer **risk score in [0, 100]** from an additive engine (`risk_engine.py`): a **base weight** for the (highest-severity) observed intent, plus **behavioural boosters** derived from session context, clamped to 100.

TABLE III -- Base intent weights.

Intent	Weight
OTHER	0
RECONNAISSANCE	10
DISCOVERY	15
CLEANUP	15
EXECUTION	20
CREDENTIAL_ACCESS	25

TABLE IV -- Behavioural boosters.

Condition	Points
Any failed authentication	+5
Brute-force pattern (>= 3 failed logins)	+10
Rapid command bursts (high command rate)	+5
Deep session (>= 5 commands)	+5
High intent variety (>= 4 distinct intents)	+5

The score maps to four **severity tiers** -- **LOW** 0-24, **MEDIUM** 25-49, **HIGH** 50-74, **CRITICAL** 75-100 -- and a session reaching **>= 75** raises an automatic **alert**. The weighting reflects kill-chain severity: exfiltration (the objective) outweighs reconnaissance (the prelude), and the boosters capture *behaviour* (brute forcing, breadth, depth) that raw intent alone misses. Because the formula is additive and transparent, every score is fully explainable -- the dashboard labels the risk "Deterministic Threat Risk" precisely because it is reproducible, not a model estimate.

VIII. Attacker Profiling

The profiler maintains two aggregation levels. A **session profile** records the intent sequence, dominant intent, furthest attack stage, command/login counts, duration, risk score, and severity -- assembling isolated events into a **multi-stage trajectory** (e.g. reconnaissance -> discovery -> credential-access -> privilege-escalation -> exfiltration -> cleanup). An **attacker profile** aggregates all sessions from one source IP: total sessions/events, peak and average risk, dominant intent, and first/last-seen timestamps -- answering whether a host is a one-off scanner or a persistent, escalating threat. These profiles back the dashboard's Session-detail and Attacker-detail views (Section IX) and the MITRE-technique overlay.

IX. Implementation

Backend. Python 3.13, FastAPI, Motor (async MongoDB). The capture->classify->score->persist path is fully asynchronous so ingestion never blocks. The REST/WebSocket surface includes health, dashboard statistics, events, sessions (with on-demand MITRE mapping and optional Gemini analysis), attackers, alerts, and six analytics endpoints (intent, severity, timeline, stage, risk-distribution, heatmap), plus `/ws/events` and `/ws/alerts` for live streaming.

Frontend. A React 19 + Vite + Tailwind + Recharts single-page analyst dashboard with five sections -- Overview, Live Sessions, Attacker Profiles, Threat Alerts, Analytics. The Overview and top navigation bar consume the two WebSockets for live, refresh-free updates; each event row shows an intent badge, severity, and a classification-source badge (with ML confidence), realising the "explainable" goal on screen. *(The companion `DASHBOARD_GUIDE.md` documents every page, term, and colour.)*

Deployment and reproducibility. The full stack -- MongoDB, Cowrie, API, and dashboard -- is containerised via Docker Compose and starts with a single command; the honeypot listens on port 2222 and the dashboard on 5173. Secrets live only in `.env` (never committed); Gemini is optional. A fixed `RANDOM_SEED = 1337` and pinned scientific-stack versions (scikit-learn 1.9.0, numpy 2.5.2) make training and evaluation deterministic and reproducible.

X. Experimental Evaluation

A. Primary evaluation (leakage-controlled held-out set)

On the 272-command held-out test set the calibrated linear SVM achieves the metrics in Table V, with per-class results in Table VI.

TABLE V -- Held-out performance (272 test commands).

Metric	Value
Accuracy	**94.49 %**
Macro Precision	0.9367
Macro Recall	0.9158
Macro F1	**0.9229**
Weighted F1	0.9436

TABLE VI -- Per-class results (held-out).

Intent	Precision	Recall	F1	Support
CLEANUP	1.000	0.941	0.970	17
CREDENTIAL_ACCESS	0.828	1.000	0.906	24
DISCOVERY	0.828	0.889	0.857	27
EXECUTION	1.000	1.000	1.000	64
EXFILTRATION	1.000	1.000	1.000	37
EXPLOITATION	1.000	1.000	1.000	20
OTHER	0.909	0.870	0.889	23
PERSISTENCE	0.970	1.000	0.985	32
PRIVILEGE_ESCALATION	1.000	0.833	0.909	12
RECONNAISSANCE	0.833	0.625	0.714	16

B. Robustness evaluation (leave-families-out)

To test generalisation to *unseen command templates*, we hold out whole families so that no test template appears in training (Table VII). This is a much harder test and the score is lower **by design**.

TABLE VII -- Primary vs. robustness evaluation.

Protocol	Accuracy	Macro-F1	Train / Test
Primary (stratified, leakage-controlled)	**94.49 %**	**0.9229**	1088 / 272
Robustness (leave-families-out)	**67.87 %**	**0.5267**	1055 / 305

C. End-to-end live verification

Beyond offline metrics, an end-to-end script drives all ten intents through the *running* system and a verifier confirms each captured command carries a classification source and, for ML/HYBRID, a real calibrated confidence -- demonstrating the ML core is genuinely in the live pipeline, not merely evaluated offline.

XI. Results and Discussion

The model classifies attacker intent with **94.49 % held-out accuracy** and a balanced **0.9229 macro-F1**, confirming that lexical features capture intent well for the majority of commands. Three classes -- EXECUTION, EXFILTRATION, EXPLOITATION -- reach perfect held-out F1, reflecting their highly distinctive vocabulary (download-and-execute chains, `tar`-and-transfer patterns, exploit strings).

The weakest class is **RECONNAISSANCE** (F1 0.714): short probes such as `whoami`, `id`, and `uname` overlap lexically with DISCOVERY and CREDENTIAL_ACCESS, and it is the smallest test class (support 16). Crucially, the confusion matrix shows errors stay *within neighbouring intents* -- no misclassification crosses into an unrelated tactic -- so mistakes are reasonable and rarely change the kill-chain interpretation. This ambiguous boundary is exactly what the rule and optional-LLM layers exist to disambiguate at runtime.

The **leave-families-out** result (67.87 %) is the honest headline for *novel* templates. A purely lexical model cannot fully generalise to command shapes it has never seen; we report this openly rather than quoting only the favourable split. It directly motivates the hybrid design: deterministic rules cover known-signature novelty, and the optional semantic LLM layer handles genuinely new commands -- so the *system's* real-world robustness exceeds that of the ML core in isolation.

Finally, the architecture delivers **explainability end-to-end**: every event exposes its deciding source and evidence, and every risk score decomposes into transparent additive terms. This auditability is a first-class property, not an afterthought -- an analyst can always answer "why was this flagged?"

XII. Limitations and Ethical / Safety Considerations

Limitations. (1) The training corpus, though realistic and family-diverse, is generated from curated templates rather than harvested from a live production honeypot; the leave-families-out result quantifies the resulting generalisation gap. (2) The classifier operates per command; richer sequence models could exploit inter-command context further. (3) Small classes (RECONNAISSANCE, PRIVILEGE_ESCALATION) have modest support and would benefit from more data.

Safety and ethics. The honeypot is strictly **isolated** and only *emulates* a shell -- **no captured attacker command is ever executed** on the host or backend; the honeypot is the sole component that handles attacker input. All demonstration attacks use **RFC-5737 documentation IP ranges** (192.0.2.0/24, 198.51.100.0/24) that route nowhere, and no real malicious payloads are run. Secrets are confined to an untracked `.env`. The system is a **defensive** research tool and should only be deployed against infrastructure one owns.

XIII. Conclusion and Future Work

We presented an intent-aware deception system that converts raw honeypot command streams into real-time, explainable attacker profiles. By fusing a deterministic rule engine, a calibrated ML core (**94.49 % held-out accuracy, 0.9229 macro-F1**), and an optional LLM semantic layer under explicit confidence arbitration -- and by scoring each session with a transparent additive risk engine -- the system infers *why* an attacker acts, assembles multi-stage trajectories, and raises scored alerts, all while keeping ML mandatory, the LLM optional, and every decision auditable. We deliberately report a stricter leave-families-out score (67.87 %) to characterise generalisation honestly.

Future work. (1) Harvest and incorporate live honeypot traffic to close the template-generalisation gap. (2) Add sequence-aware modelling over the command stream to capture inter-command context. (3) Expand the rule/pattern library and the intent taxonomy. (4) Study adversarial robustness to command obfuscation. (5) Explore automated adaptive deception -- having the honeypot respond differently based on inferred intent.

References

(IEEE style; fill in exact page numbers/DOIs as needed for your venue.)

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Appendix A -- Suggested figures to render for the paper

1. **System architecture diagram** (from Section III -- redraw as a clean vector figure).
2. **Confusion matrix heatmap** -- available at `docs/evaluation/confusion_matrix.svg`.
3. **Per-class F1 bar chart** -- `docs/evaluation/per_class_f1.svg`.
4. **Hybrid arbitration flowchart** (from Section VI).
5. **A live dashboard screenshot** showing intent badges, risk scores, and source badges.

Appendix B -- Reproducibility checklist

- Fixed seed `RANDOM_SEED = 1337`; deterministic dataset generation, split, CV, and training.
- Environment: Python 3.13.12, scikit-learn 1.9.0, numpy 2.5.2.
- Leakage control asserted in code (no shared command string across train/test).
- Both favourable (stratified) and conservative (leave-families-out) scores reported.
- Metrics regenerated by re-running the training script; nothing hand-edited.